

# Stochastic Mathematics for Finance

## Week 2 Assignment

### Mathematical and Computational Case Studies in Stochastic Finance

#### Instructions

- This assignment contains 4 computational probability and Monte Carlo based case studies.
- Each question contains both a theoretical component and a computational component.
- For computational parts, submit the relevant Python code and a short explanation of your observations.
- You are encouraged to visualize your simulations wherever appropriate.
- Briefly interpret the meaning of your numerical and simulation results.
- You may use the Week 2 resources, but some parts require applying the ideas in a new setting.

#### Submission Format

Submit your work on GitHub in a clean repository containing:

- a `README.md` file with your name, roll number, and GitHub username;
- a single PDF named `solutions.pdf`;
- Python code files inside a folder named `code/`, named clearly as `q1.py`, `q2.py`, etc.;
- plots or generated outputs, if any, inside a folder named `outputs/`.

**Question 1: Empirical Central Limit Theorem**

A quantitative analyst is studying how aggregated random market shocks behave as the number of shocks increases.

Suppose:

$$X_1, X_2, \dots, X_n$$

are iid Bernoulli random variables satisfying:

$$P(X_i = 1) = 0.5, \quad P(X_i = 0) = 0.5.$$

Define:

$$S_n = X_1 + \dots + X_n.$$

The standardized sum is:

$$Z_n = \frac{S_n - n/2}{\sqrt{n/4}}.$$

(a) Compute:

$$E[S_n] \quad \text{and} \quad \text{Var}(S_n).$$

(b) Explain why the standardization above produces a variable with approximately mean 0 and variance 1.

(c) Using Python, simulate 10000 realizations of  $Z_n$  for:

$$n = 10, 50, 200.$$

(d) Plot histograms of the simulated standardized variables.

(e) Overlay the standard Gaussian density on each histogram.

(f) Briefly explain how the shape of the distribution changes as  $n$  increases.

(g) Estimate the empirical mean and variance of  $Z_n$  for each value of  $n$ .

(h) Explain how this experiment illustrates the Central Limit Theorem.

**Hint**

You may use:

$$\text{Var}(X_i) = 0.25.$$

For plotting the Gaussian density, the function `scipy.stats.norm.pdf` may be useful.

**Question 2: Monte Carlo Integration**

A quantitative researcher wants to estimate integrals using random sampling instead of deterministic numerical methods.

Consider:

$$I = \int_0^1 e^{-x^2} dx.$$

(a) Explain why:

$$I = E[e^{-U^2}],$$

where:

$$U \sim \text{Uniform}(0, 1).$$

(b) Using Python, estimate  $I$  using Monte Carlo simulation with:

$$N = 10^2, 10^3, 10^4, 10^5.$$

- (c) For each value of  $N$ , compute the absolute error relative to the true numerical value of the integral.
- (d) Plot the Monte Carlo estimate as a function of  $N$ .
- (e) Briefly explain why the approximation improves as  $N$  increases.
- (f) Explain why Monte Carlo integration is especially useful in high-dimensional problems.

**Hint**

You may use:

$$\frac{1}{N} \sum_{i=1}^N e^{-U_i^2}$$

as your estimator.

You may use `scipy.integrate.quad` to compute the true numerical value of the integral.

**Question 3: Monte Carlo Estimation of  $\pi$** 

A computational finance team is testing random sampling methods by estimating geometric quantities probabilistically.

Consider the square:

$$[-1, 1] \times [-1, 1].$$

Inside the square lies the unit circle:

$$x^2 + y^2 \leq 1.$$

- (a) Explain why the probability that a uniformly sampled point lies inside the circle is:

$$\frac{\pi}{4}.$$

- (b) Using Python, generate  $N$  random points uniformly in the square for:

$$N = 10^2, 10^3, 10^4, 10^5.$$

- (c) Estimate  $\pi$  using:

$$\pi \approx 4 \times (\text{fraction of points inside the circle}).$$

- (d) Plot the randomly generated points for one simulation, clearly distinguishing points:

- inside the circle;
- outside the circle.

- (e) Plot the estimated value of  $\pi$  as a function of  $N$ .

- (f) Briefly explain why the estimate becomes more accurate as  $N$  increases.

- (g) Compare the computational estimate with the true value:

$$\pi \approx 3.14159.$$

**Hint**

For each sampled point:

$$(x, y),$$

check whether:

$$x^2 + y^2 \leq 1.$$

**Question 4: Monte Carlo Option Pricing**

A quantitative trading desk models the future stock price using a simple stochastic model.

Suppose the terminal stock price is modelled as:

$$S_T = S_0 e^{\sigma Z},$$

where:

$$Z \sim \mathcal{N}(0, 1).$$

Assume:

$$S_0 = 100, \quad \sigma = 0.2, \quad K = 110.$$

A European call option has payoff:

$$(S_T - K)^+,$$

where:

$$(x)^+ = \max(x, 0).$$

- (a) Explain why the option payoff is itself a random variable.
- (b) Using Python, simulate 100000 realizations of:

$$S_T.$$

- (c) Compute the simulated call option payoffs.
- (d) Estimate:

$$E[(S_T - K)^+]$$

using Monte Carlo simulation.

- (e) Plot a histogram of the simulated terminal stock prices.
- (f) Plot a histogram of the simulated option payoffs.
- (g) Repeat the simulation for:

$$\sigma = 0.1, \quad \sigma = 0.3, \quad \sigma = 0.5.$$

- (h) Briefly explain how volatility affects:

- the spread of terminal stock prices;
- the expected option payoff.

- (i) Explain why Monte Carlo methods are useful in derivative pricing.

**Hint**

You may use:

$$\text{np.maximum}(S_T - K, 0)$$

to compute the option payoff.

## **General Notes**

Throughout this assignment, focus on:

- understanding simulation behaviour;
- interpreting convergence empirically;
- comparing theoretical quantities with numerical estimates;
- visualizing probabilistic phenomena computationally.

Many modern quantitative finance systems rely heavily on simulation and Monte Carlo methods. This assignment is intended to provide intuition for how probabilistic models behave in practice.